

Mid-term exam on Compilers

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Question 1. Let the alphabet $\Sigma = \{a, b\}$ and the following regular expressions:

$$r = a(a|b)^*ba$$

$$s = (ab)^*|(ba)^*|(a^*|b^*)$$

The language denoted by r is noted $L(r)$ and the language denoted by s is noted $L(s)$.

Find a word x such as

1. $x \in L(r)$ and $x \notin L(s)$
2. $x \notin L(r)$ and $x \in L(s)$
3. $x \in L(r)$ and $x \in L(s)$
4. $x \notin L(r)$ and $x \notin L(s)$

Question 2. Given the binary alphabet $\Sigma = \{a, b, c\}$ and the order on letters $a < b < c$, write regular definitions for the following languages.

1. All words starting and ending with c .
2. All words in which the third last letter is b .
3. All words containing exactly three a .
4. All words containing at least one b before a c .
5. All words in which the letters are in increasing order.

Question 3. Simplify, if possible, the following regular expressions.

$$\begin{aligned} &(\epsilon|a^*|b^*|a|b)^* \\ &a(a|b)^*b|(ab)^*|(ba)^* \end{aligned}$$